HAlg Exam Sheet 2025/26 - What Doesn't Kill You

Theorem: Random Theorems And Definition

- $\mathrm{pr}_i : X_1 \times \ldots \times X_n \to X_i, (x_1,\ldots,x_n) \mapsto x_i$
- $\mathcal{P}(X) = \{U : U \subseteq X\}$ is the *power set* of X. A subset of $\mathcal{P}$ is a *system of subsets* of X.

Remark: Notation

- 'almost all' means 'all but finitely many'

Vector Spaces

Definition: Systems Of Linear Equations

Let F be a field. We seek to solve the system of n linear equations in m unknowns, described more easily by the extended coefficient matrix.

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1m}x_m &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2m}x_m &= b_2 \\ \vdots \qquad \vdots \qquad \vdots \qquad &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nm}x_m &= b_n \end{aligned} \qquad \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1m} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2m} & b_2 \\ \vdots & & & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} & b_n \end{array}\right)$$

- The set $L \subseteq F^m$ of m-tuples satisfying the equations is the *solution set*.
- If the solution set of a linsys of eqns $\neq \emptyset$ we obtain all solutions by adding componentwise an arbitrary solution of the homogenised system to a fixed solution of the system.

Definition: Matrices

A mapping $A : \{1,\ldots,n\} \times \{1,\ldots,m\} \to Z$ is called an $n \times m$-matrix with coefficients in Z.
The set of all $n \times m$-matrices with coefficients in Z is denoted $\mathrm{Mat}(n \times m; Z) := \mathrm{Maps}(1,\ldots,n \times 1,\ldots,m, Z)$
If $m = n$ we have a *square matrix* and write $\mathrm{Mat}(n, Z)$

Definition: Fields And Vector Spaces

A *field* F is a set with functions $+ : F \times F \to F; (\lambda,\mu) \mapsto \lambda + \mu$ and $\cdot : F \times F \to F; (\lambda,\mu) \mapsto \lambda\mu$ s.t. $(F,+)$ and $(F \setminus \{0\}, \cdot)$ are abelian groups with respective identities $0_F, 1_F$, and $\lambda(\mu + \nu) = \lambda\mu + \lambda\nu \in F$.

- $0\vec{v} = \vec{0}; (-1)\vec{v} = -\vec{v}$
- $\forall \lambda \in F,\ \lambda\vec{0} = \vec{0}$
- If $\lambda\vec{v} = 0, \lambda = 0$ or $\vec{v} = \vec{0}$

A *vector space* V *over a field* F is a pair made up of an abelian group $V = (V, \dot{+})$ and a mapping $F \times V \to V; (\lambda, \vec{v}) \mapsto \lambda\vec{v}$ such that for all $\lambda,\mu \in F$ and $\vec{v},\vec{w} \in V$ the following hold:

$$\lambda(\vec{v} \dot{+} \vec{w}) = (\lambda\vec{v}) \dot{+} (\lambda\vec{w})$$
$$(\lambda + \mu)\vec{v} = (\lambda\vec{v}) \dot{+} (\mu\vec{v})$$
$$\lambda(\mu\vec{v}) = (\lambda\mu)\vec{v}$$
$$1_F\vec{v} = \vec{v}$$

- Let X be a set and V an $F-$VS. Then the set $\mathrm{Maps}(X,V)$ of mappings $X \to V$ is an $F-$VS if we define $(f+g)(x) = f(x) + g(x)$ and $(\lambda f)(x) = \lambda(f(x))$.
- Given $F-$VSs $V_1,\ldots,V_n$, $V_1 \times \ldots \times V_n$ is an $F-$VS if we define addition/mult by scalars componentwise. We denote this VS by $V_1 \oplus \ldots \oplus V_n$ and name it the *external direct sum*. $F^n = F \oplus \ldots \oplus F$ n times.

Definition: Vector Subspaces

A subset U of a vector space V is a *vector subspace* if $\vec{0} \in U$ and $\forall \vec{u},\vec{v} \in U, \lambda \in F$, $\vec{u} + \vec{v}$ and $\lambda\vec{u} \in U$.

- The solution set of a linsys of eqns seen earlier is a vector space
- An expression of form $\alpha_1\vec{v}_1 + \ldots + \alpha_n\vec{v}_n$ with n finite is a *linear combination* of $\vec{v}_1,\ldots,\vec{v}_n$.
- A subset is a *hyperplane* if it is a proper subspace and together with one other $\vec{v}$ generates the whole VS.
- The intersection of two VSs is itself a VS

Let T be a subset of V, a VS over F. $\langle T \rangle$, the *span of* T or *vector subspace generated by* T is the set of all linear combinations of vectors in T. (If $T = \emptyset, \langle T \rangle = \vec{0}$). Note if $v \in \langle T \rangle, \langle T \cup \{v\} \rangle = \langle T \rangle$.
A subset of V is a *generating/spanning set* if its span is V. A VS with a finite generating set is *finitely generated*.

Definition: Linear Independence

A subset L of a vector space V is *linearly independent* if for all $\alpha_1,\ldots,\alpha_n \in F$ and distinct $\vec{v}_1,\ldots,\vec{v}_n \in L$, $\alpha_1\vec{v}_1 + \ldots + \alpha_n\vec{v}_n = 0$ only if all $\alpha_i = 0$. Otherwise it is *linearly dependent*.

Linearly Independent	Linearly Dependent
$\emptyset; \{\vec{v}\}, \vec{v} \neq \vec{0}$	$\{\vec{0}\}$

Definition: Bases

Let V be a VS over a field F. A *basis* of V is a linearly independent generating set in V.

- The family $(\vec{v}_i)_{1 \leq i \leq n}$ is a basis of V iff $\Phi : F^n \to V; (\alpha_1,\ldots,\alpha_n) \mapsto (\alpha_1\vec{v}_1 + \ldots + \alpha_n\vec{v}_n)$ is a bijection.
- TFAE for a subset E of a VS V:
  - (i) E is a basis
  - (ii) E is minimal among generating sets ($\forall \vec{v} \in E$, $E \setminus \vec{v}$ does not generate V)
  - (iii) E is maximal among linearly independent subsets ($\forall \vec{v} \in E, E \cup \vec{v}$ is linearly dependent)
- If V is a finitely generated VS, it has a finite basis
- For a set X, the set $\mathrm{Maps}(X,F)$ equipped with pointwise addition and scalar multiplication is an F-VS. The subset, $F\langle X \rangle \subseteq \mathrm{Maps}(X,F)$, of all mappings which send almost all $x \in X$ to zero as a vector subspace, called the *free vector space on* X
- TFAE
  - (i) The family $(\vec{v}_i)_{i \in I}$ is a basis of V.
  - (ii) For all $v \in V$, $\exists$ precisely one family $(a_i)_{i \in I}$ s.t. almost all $a_i = 0$, and $\vec{v} = \sum_{i \in I} a_i\vec{v}_i$.

Theorem: FTLA, Steinitz, Exchange Lemma

Let V be a VS, $L \subset V$ linearly ind., and $E \subseteq V$ generating

- If V is finitely generated then it has a finite basis, and any two bases have the same cardinality
- Fundamental Theorem of Linear Algebra: $|L| \leq |E|$.
- Steinitz Exchange Theorem: There is an injection $\phi : L \hookrightarrow E$ s.t. $\{E \setminus \phi(L)\} \cup L$ is also generating.
- Exchange Lemma: Steinitz but for single elements.

Definition: Dimensions Of Vector Spaces

Let V be a finitely-generated VS, $L \subset V$ linearly ind., $E \subseteq V$ generating. For finitely-generated V, its *dimension*, $\dim V$ is the cardinality of a basis of V. For dimension as an F-VS write $\dim_F V$. Also:

- $|L| \leq \dim V$, if $|L| = \dim V$ then L is a basis
- $|E| \geq \dim V$, if $|E| = \dim V$ then E is a basis
- A proper vector subspace of V has a strictly smaller dimension than V
- For U,W subspaces of V, $\dim(U+W) + \dim(U \cap W) = \dim(U) + \dim(W)$.
- Given $V_1,\ldots,V_n$ $F-$VSs, $\dim(V_1 \oplus \ldots \oplus V_n) = \dim(V_1) + \ldots + \dim(V_n)$

MDfn: Linear Mappings: Homo/Iso/Endo/Automorphisms

Let V,W be VSs over a field F and $B \subset V$ be a basis. A mapping $f : V \to W$ is *(F-)linear* or a *homomorphism of F-vector spaces* if for all $\vec{v}_1,\vec{v}_2 \in V$ and $\lambda \in F$: $F(\vec{v}_1 + \vec{v}_2) = f(v_1) + f(v_2)$ and $f(\lambda\vec{v}_1) = \lambda f(\vec{v}_1)$. The set of all homomorphisms from V to W is $\mathrm{Hom}_F(V,W) = \mathrm{Hom}(V,W) \subseteq \mathrm{Maps}(V,W)$

- An *isomorphism* is a bijective linear mapping. Isomorphisms are invertible, and the inverse mapping is also an isomorphism
- An *endomorphism* is a homomorphism from a VS to itself
- An *automorphism* is an isomorphism from a VS to itself. The automorphisms of a VS V form a subgroup of its permutation group called the *general linear* or *automorphism* group, denoted $GL(V) = \mathrm{Aut}(V) \subseteq Maps^{\times}(V,V)$.
- If $f : V \to W$ and $g : U \to V$ are homo.s, $(f \circ g) : U \to W$ is a homo.
- A homo. can be restricted to a bijection $\mathrm{Hom}_F(V,W) \twoheadrightarrow \mathrm{Maps}(B,W)$, $f \mapsto f|_B$ - ie a linear mapping determines and is determined by the values it takes on a basis.
- Every injective homo. $f : V \hookrightarrow W$ has a *left inverse* g s.t. $g \circ f = \mathrm{id}_V$
- Every surjective homo. $f : V \twoheadrightarrow W$ has a *right inverse* g s.t. $f \circ g = \mathrm{id}_W$
- For $n \geq 0 \in \mathbb{Z}$, a VS V over F is isomorphic to F^n iff $\dim V = n$.
- A point sent to itself by a homo. is a *fixed point* of the mapping. Given a mapping $f : X \to X$, the set of fixed points is $X^f = \{x \in X : f(x) = x\}$.

Definition: Image and Kernel

Let V,W be VSs and $f : V \to W$ be a linear mapping. The *image* of f is $(f) = F(V) \subseteq W$. It is a VsubS of W.
The *kernel* of f is $\ker(f) := f^{-1}(\vec{0}) = \{\vec{v} \in V : f(\vec{v}) = \vec{0}\}$. It is a VsubS of V.

- A linear mapping f is injective iff its kernel is $\vec{0}$.
- The image of a VsubS under a linear mapping is again a VsubS, and the preimage of a VsubS under a linear mapping is again a VsubS.

Theorem: Rank-Nullity Theorem

Let V,W be VSs and $f : V \to W$ a linear mapping.
Then $\dim V = \dim(\ker(f)) + \dim(\mathrm{im}(f))$.
$\dim(\mathrm{im}(f))/\dim(\ker(f))$ are the *rank/nullity* of f.

Linear Mappings and Matrices

Theorem: Linking Linear Mappings And Matrices

Let F be a field and $m,n \in \mathbb{N}$. There is a bijection M between the space of linear mappings $F^m \to F^n$ and $\mathrm{Mat}(n \times m; F)$ which maps $f \mapsto [f]$.
$M(f) := [f]$ is the *representing matrix* of f, whose columns are the images under f of the standard basis elements of F^m, ie $[f] = (f(\vec{e}_1)|\ldots|f(\vec{e}_m))$.

Theorem: Composing Mappings vs Matrix Multiplication

Let $g : F^\ell \to F^m$ and $f : F^m \to F^n$ be linmaps. The representing mat of their composition is the product of their representing mats: $[f \circ g] = [f][g]$.

- For $\ell,m,n \in \mathbb{N}$, F a field, $A \in \mathrm{Mat}(n \times m; F)$, $B \in \mathrm{Mat}(m \times \ell; F)$. Then $AB \in \mathrm{Mat}(n \times \ell)$ is defined by $(AB)_{ik} = \sum_{j=1}^m A_{ij}Bjk$.

Definition: The General Linear Group

The set of invertible $n \times n$ matrices in a field F form a group under matrix multiplication, called the *General Linear group of* $(n \times n)$ *matrices* and written $\mathrm{GL}(n;F) := \mathrm{Mat}(n;F)^{\times}$

Definition: Properties Of Matrices

- Let $k,\ell,m,n \in \mathbb{N}$, $A,A' \in \mathrm{Mat}(n \times m; F)$, $B,B' \in \mathrm{Mat}(m \times \ell; F)$, $C \in \mathrm{Mat}(\ell \times k; F)$, and $I = I_m$. Then:
  - (i) $(A + A')B = AB + A'B$
  - (ii) $A(B + B') = AB + AB'$
  - (iii) $(AB)C = A(BC)$
- A square matrix A is *invertible* if $\exists A^{-1}$ of same size s.t $AA^{-1} = A^{-1}A = I$
- An *elementary* matrix is one which differs from I in at most one entry
- Every square matrix with entries in a field can be written as a product of elementary matrices
- A matrix whose only non-zero entries lie on the diagonal, and which has first ones along the diagonal, then zeroes, is in *Smith Normal Form*. For each $A \in \mathrm{Mat}(n \times m; F)$, $\exists$ invertible P,Q s.t. PAQ is in Smith Normal Form.
- The *rank* of a matrix A is the dimension of the subspace of F^n generated by the columns or rows of A (the *dim* of each are always equal).
- A is of *full rank* if the rank is equal to the number of rows or columns
- $\mathrm{rk}(A+B) \leq \mathrm{rk}A + \mathrm{rk}B$, and $\mathrm{rk}(AB) \leq \min(\mathrm{rk}A, \mathrm{rk}B)$.